

Madhu Babu Yalavarthi

Full stack developer

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SUMMARY

- Full Stack Developer with 6+ years of experience designing scalable, high-performance web applications, specializing in Python, Django, FastAPI, React.js, and TypeScript across e-commerce and SaaS platforms.
- Proven expertise in RESTful APIs, GraphQL, and microservices architecture, enabling seamless third-party integrations, optimized merchant dashboards, and enhanced developer collaboration tools.
- Strong background in cloud and DevOps with AWS, Azure, GCP, Docker, Kubernetes, and CI/CD pipelines, improving deployment efficiency, system scalability, and application uptime in production environments.
- Hands-on experience implementing AI-powered search, recommendations, and real-time analytics using PyTorch, TensorFlow, Pandas, and PySpark, driving measurable improvements in engagement and decision-making.
- Collaborative team player with experience in mentorship, Agile/Scrum environments, and cross-functional development, consistently delivering business-impacting features while maintaining 70%+ test coverage and code quality.

PROFESSIONAL EXPERIENCE

SHOPIFY, San Francisco, CA

07-2023 to Present

Full stack Developer

- Engineered scalable e-commerce modules using Python, Django, and FastAPI, enabling faster product catalog updates and improving merchant onboarding efficiency by 20% across Shopify Plus clients.
- Designed and optimized RESTful APIs and GraphQL services powering merchant dashboards, reducing API response times by 15% and ensuring seamless integrations with third-party payment gateways.
- Built dynamic, responsive UIs using React.js, TypeScript, and Redux, improving customer checkout experience and increasing conversion rates by 12% across enterprise storefronts.
- Developed AI-powered personalized storefront search and recommendation features using Python FastAPI, PyTorch, and TensorFlow, boosting product discovery and increasing customer engagement by 10%.
- Collaborated with data teams to implement ETL pipelines with Python, Pandas, and PySpark, enabling real-time sales analytics and boosting reporting accuracy by 15% for merchants.
- Automated deployment pipelines with Docker, Kubernetes, and Jenkins, reducing release times by 20% and ensuring high availability of production environments.
- Integrated Azure services (Functions, Virtual Machines, Blob Storage, SQL Database) into core applications, improving scalability and achieving 99% uptime across high-traffic merchant-facing applications.
- Applied unit testing and integration testing with Pytest and Cypress, increasing test coverage to 70% and improving overall product release reliability.
- Implemented CI/CD pipelines with GitHub Actions, enabling faster feature rollouts and reducing deployment-related incidents by 15%.
- Optimized SQL and NoSQL database queries in PostgreSQL and MongoDB, enhancing transaction processing speed by 12% during peak sales events like Black Friday.
- Integrated AI-driven recommendation APIs into React.js and Next.js storefronts, enabling real-time personalized product suggestions and increasing average order value by 10% across merchant stores.
- Mentored junior developers in full stack best practices, debugging, and system design principles, improving team efficiency and accelerating delivery of customer-facing features.

POSTMAN, Bangalore, IND

07-2019 to 11-2022

Full stack developer

- Designed and maintained scalable backend services using Python, Django, and Flask, improving API performance by 15% and supporting seamless integration with Postman's API collaboration platform.
- Developed RESTful APIs and microservices for automation features, enabling faster request handling and reducing API response latency by 10% across millions of global developer users.
- Built Python-based automation scripts for regression testing and CI/CD workflows, reducing manual testing efforts by 20% and accelerating release cycles with Jenkins and GitHub Actions.
- Partnered with product teams to deliver backend APIs for monitoring and collaboration dashboards, enhancing developer productivity and improving Postman's user engagement by 12%.
- Developed ETL pipelines with Python, Pandas, and PySpark, automating processing of API usage data and generating insights that improved customer adoption of advanced platform features.
- Implemented robust unit testing frameworks with Pytest and Unittest, achieving 70%+ code coverage and ensuring stability of new releases before deployment in production environments.
- Optimized PostgreSQL and MongoDB queries within backend services, reducing query execution times by 15% and enabling real-time API performance tracking across enterprise clients.
- Containerized Python microservices using Docker and Kubernetes, ensuring scalable deployments and reducing infrastructure-related downtime by 15% across staging and production environments.
- Secured APIs by implementing OAuth 2.0 and JWT authentication mechanisms, strengthening compliance with enterprise-grade data security and protecting sensitive API credentials.
- Collaborated with senior developers on code reviews, debugging, and Python best practices, contributing to team-wide improvements in maintainability, scalability, and overall code quality.

SKILLS

Languages & Frameworks: Python, JavaScript (ES6+), TypeScript, React, Next.js, Node.js, Express, Django, GraphQL.

Frontend & UI: HTML5, CSS3 (Flexbox, Grid), component-based design, responsive layouts, cross-browser compatibility, modern bundlers (Webpack, Vite, Babel), Figma for design collaboration, performance optimizations (code splitting, Core Web Vitals).

Backend & Server-Side: RESTful API design, GraphQL, WebSockets (Socket.io), server-side rendering, microservices architecture, Redis caching, PostgreSQL and MySQL optimization, Flask, authentication/authorization workflows, JSON data interchange.

Databases & Caching: PostgreSQL, MySQL, MongoDB (NoSQL), Redis; skilled in SQL optimization, indexing, ACID transactions, data classification, choosing DB paradigms.

Cloud & DevOps: AWS (EKS, Lambda, RDS, CloudFormation), GCP (App Engine, Cloud SQL, GCS), Firebase, Docker, Kubernetes, Terraform, Ansible, Bash scripting, Linux administration, serverless architectures (Lambda or Cloud Functions), CI/CD pipelines (GitHub Actions, Jenkins).

Testing & QA: Unit, integration, and test-driven development using Jest, Mocha, JUnit, Cypress, Postman; experience with automated testing frameworks and maintaining regression coverage.

Security & Best Practices: OAuth 2.0, JWT, RBAC, OWASP secure coding, input validation, software security best practices, XSS or SQL injection prevention.

Version Control & Collaboration: Git, GitHub, GitLab, Agile (Scrum or Kanban), Jira, Confluence; effective teamwork, code reviews, cross-functional collaboration, and mentoring.

Soft Skills & Problem Solving: Critical thinking, advanced analytics, adaptability, entrepreneurial mindset, effective verbal and written communication, project management, collaboration, and time management for navigating complex and fast-changing

EDUCATION

Masters in Computer Programming
Youngstown State University